

We now take up the question of how to find the ‘volume under a surface’ if your domain is *not* a simple rectangle. Note that arbitrary regions R can get *really* weird looking! For the most part, however, we will consider ‘general’ regions to be those like polygons, ellipses, and closed smooth curves. For any of these regions, we still have

$$\text{Volume} = \iint_R f(x, y) dA,$$

where $z = f(x, y)$ is the surface in question. However, evaluating this difficult for general R is rather difficult. What do we choose for our integrating bounds?

Sometimes our region R is ‘simple’ enough. For example, if our region R looks like it *was* a rectangle with $a \leq x \leq b$, but now the ‘top’ and ‘bottom’ bounds for y are not horizontal lines—suppose they are now nice, simple, continuous functions $g_1(x)$ and $g_2(x)$:

We see that it is still possible to compute the volume:

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

In this case, we say that R is *vertically simple* (i.e., we will use *vertical cross-sections*).

Definition. Let R be a closed and bounded region in the xy -plane. Then we say that R is:

- **Vertically simple** (y -simple) if there are functions $g_1(x)$ and $g_2(x)$ and constants a and b such that for each point (x, y) in the region R we have $g_1(x) \leq y \leq g_2(x)$ and $a \leq x \leq b$.
- **Horizontally simple** (x -simple) if there are functions $h_1(y)$ and $h_2(y)$ and constants c and d such that for each point (x, y) in the region R we have $h_1(y) \leq x \leq h_2(y)$ and $c \leq y \leq d$.

Theorem. [Fubini's Theorem (Simple Form)] Let $z = f(x, y)$ be continuous on a region R .

(i) If R is y -simple, then
$$\iint_R f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx.$$

(ii) If R is x -simple, then
$$\iint_R f(x, y) \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy.$$

Example. Consider the region R in the xy -plane bounded by the x -axis, $y = x$, and $x = 1$.

(i) Draw the region R and describe its 'simple' structure.

(ii) Find the volume under the surface $f(x, y) = 3 - x - y$ over the region R .

(iii) Find the volume under the surface $f(x, y) = \frac{\sin(x)}{x}$ over the region R .

What the last example showcased is that changing the order of integration is not only useful, but also *necessary* sometimes. Therefore, we need to develop the skill of switching the order on the fly – but this can be tricky! We will need to change the integration bounds and this usually requires a picture.

Example. Consider the following double integral: $\int_0^2 \int_{x^2}^{2x} 4x + 2 \, dy \, dx$. Write an equivalent integral with the order of integration reversed. Evaluate both integrals and compare their values.

Example. Find the volume under the surface $f(x, y) = 16 - x^2 - y^2$ over the region R bounded by the curve $y = 2\sqrt{x}$, the line $y = 4x - 2$, and the x -axis.